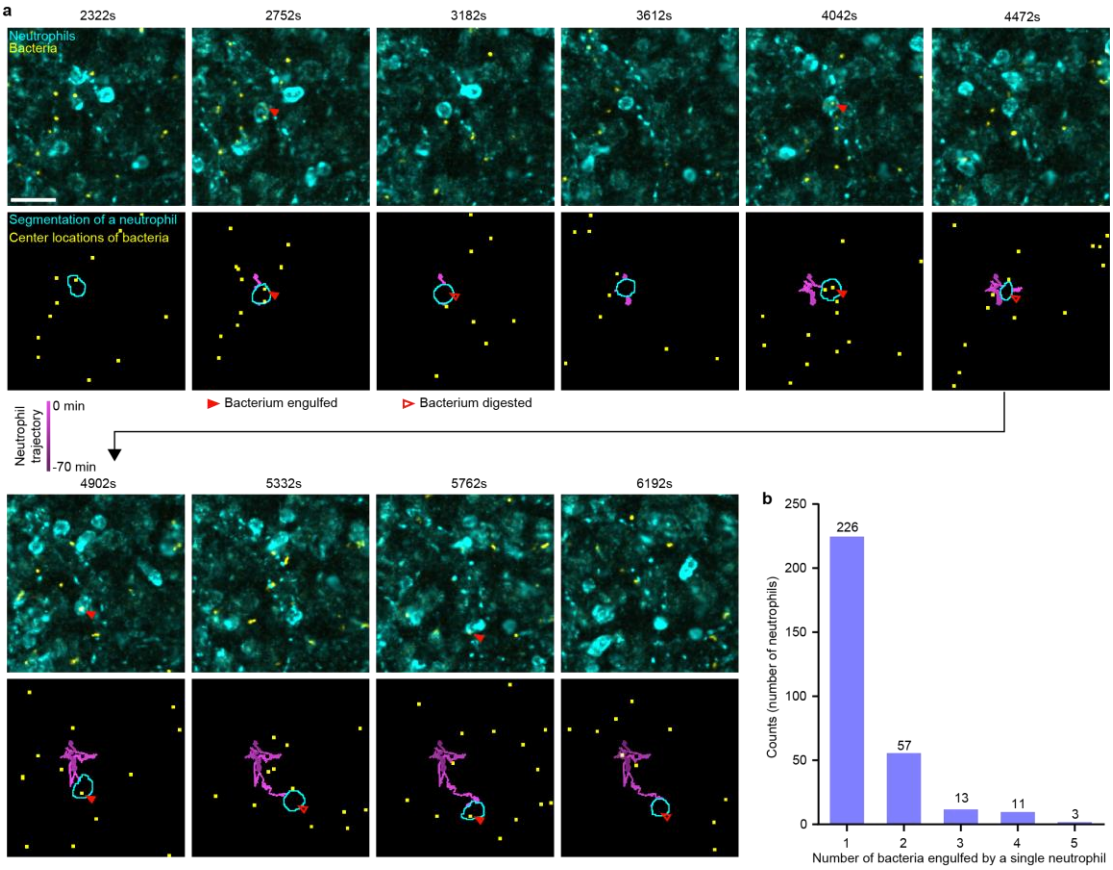


Supplementary Figure 20 | Continuous tracking and segmentation of single neutrophil chasing and engulfing bacteria. a, The upper row shows a 2-channel sequential image dataset of bacteria and neutrophils over a 70-minute time window, while the lower row shows representative tracking and segmentation of a single neutrophil pursuing bacteria. The segmented neutrophil is outlined with a solid cyan line, and trajectory is shown as a purple gradient line. Bacterial center positions are marked yellow. Engulfment and digestion events are indicated by red-filled and red-outline triangles, respectively. **b**, Histogram of the number of bacterial engulfment events per neutrophil. The data used here were acquired via 2pSAM imaging of neutrophils, macrophages, and bacteria in mouse spleen, consistent with the dataset shown in **Fig. 4**. CELLECT results were generated using the model trained on sequence #3 of the mskcc-confocal dataset. Scale bars: 20 μm